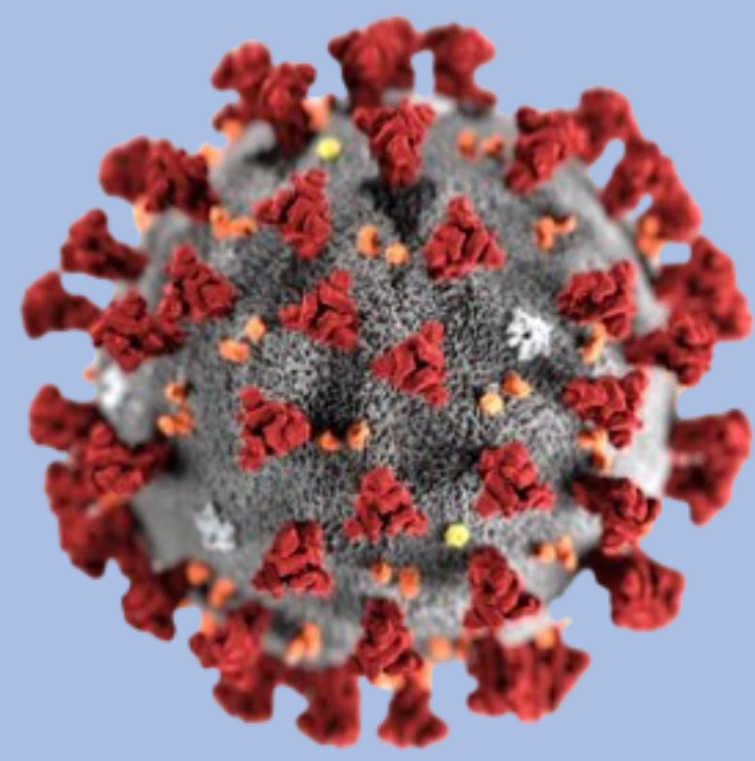




Lost Connection: How the Digital Divide Impacted Student Achievement in the Wake of the COVID-19 Pandemic

By Patrick Rourke | SOC 1340 | December 18, 2024



Introduction

During the COVID-19 pandemic, schools across the country shifted to a remote learning model in which students attended classes virtually and completed asynchronous homework assignments. As a result of the lack of in person instruction, test scores dropped. While great strides have been made to close the digital divide in the United States, some sources estimate that more than 12% of households do not have access to the internet. During the pandemic, students living in these households would not have been able to attend online classes and would have missed out on crucial learning opportunities.

Research Question

Were decreases in Math and ELA proficiency from before to after the pandemic more significant in school districts with higher percentages of people with no access to the internet?

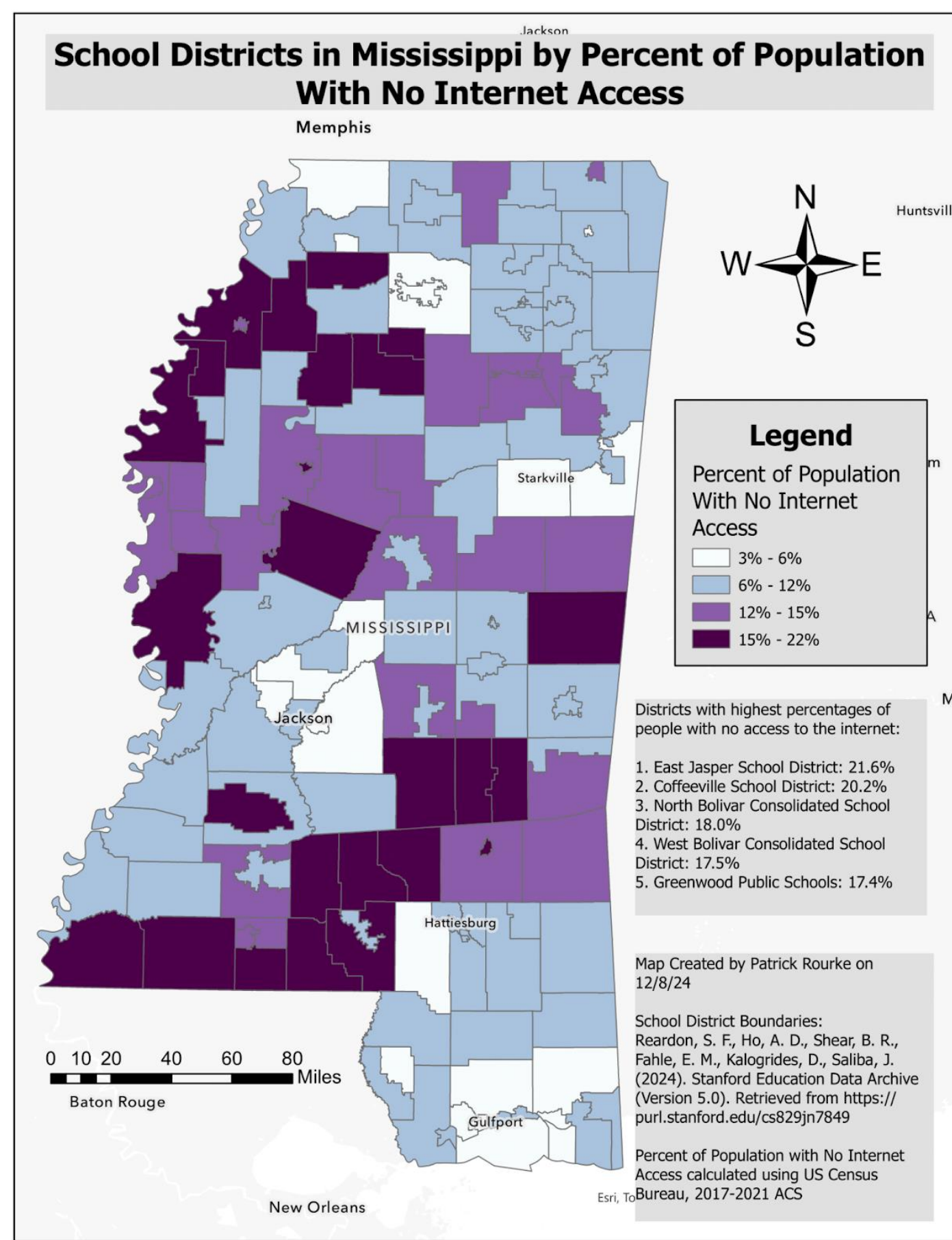
Why These States?

These are two very different states and make for a good comparison. Mississippi: South, Republican, high percentage of population with no internet access
Washington: Pacific Northwest, Democratic, low percentage of population with no internet access

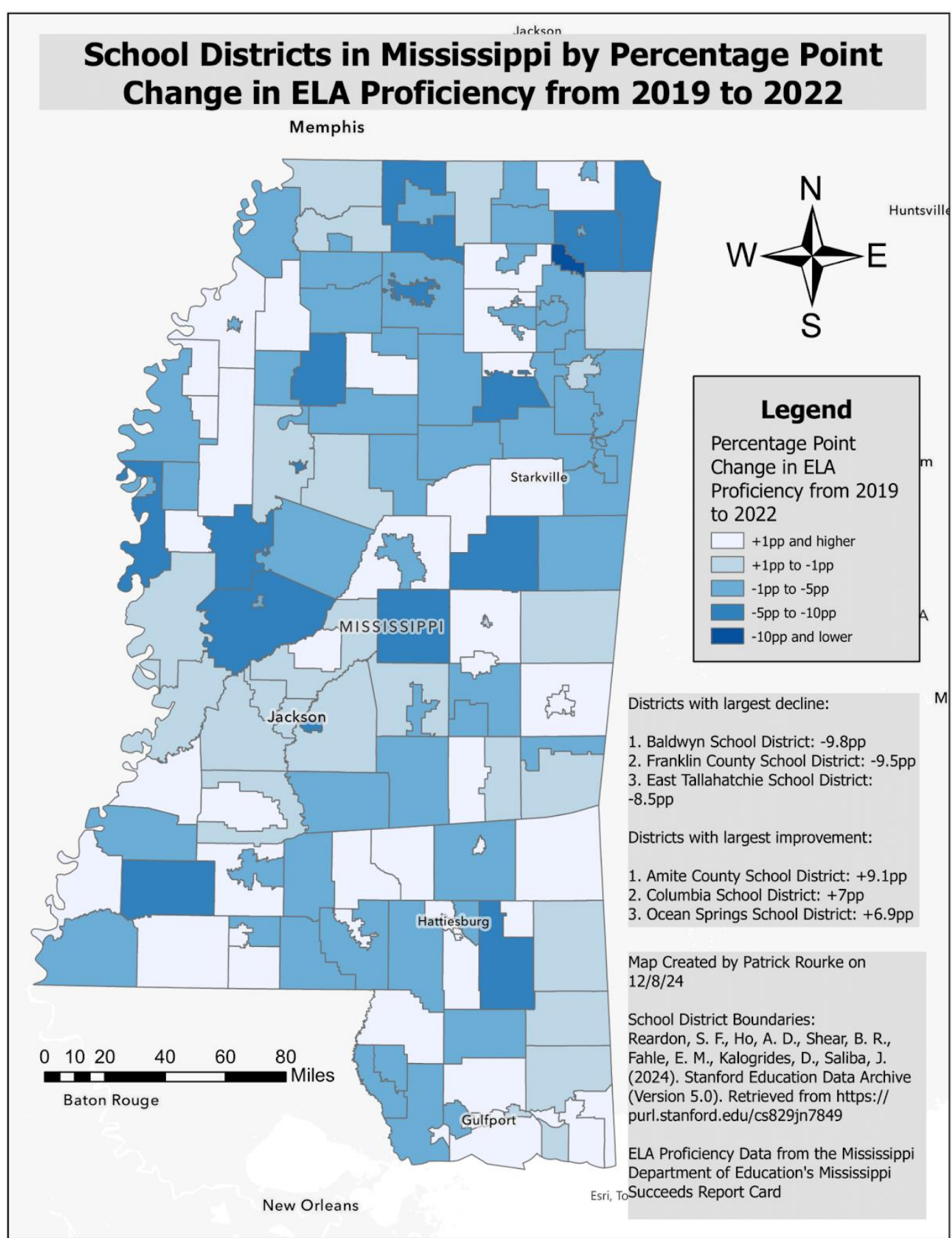
Why ELA and Math?

- These are the two most commonly assessed subjects on standardized tests.
- All states set performance standards for these two areas.

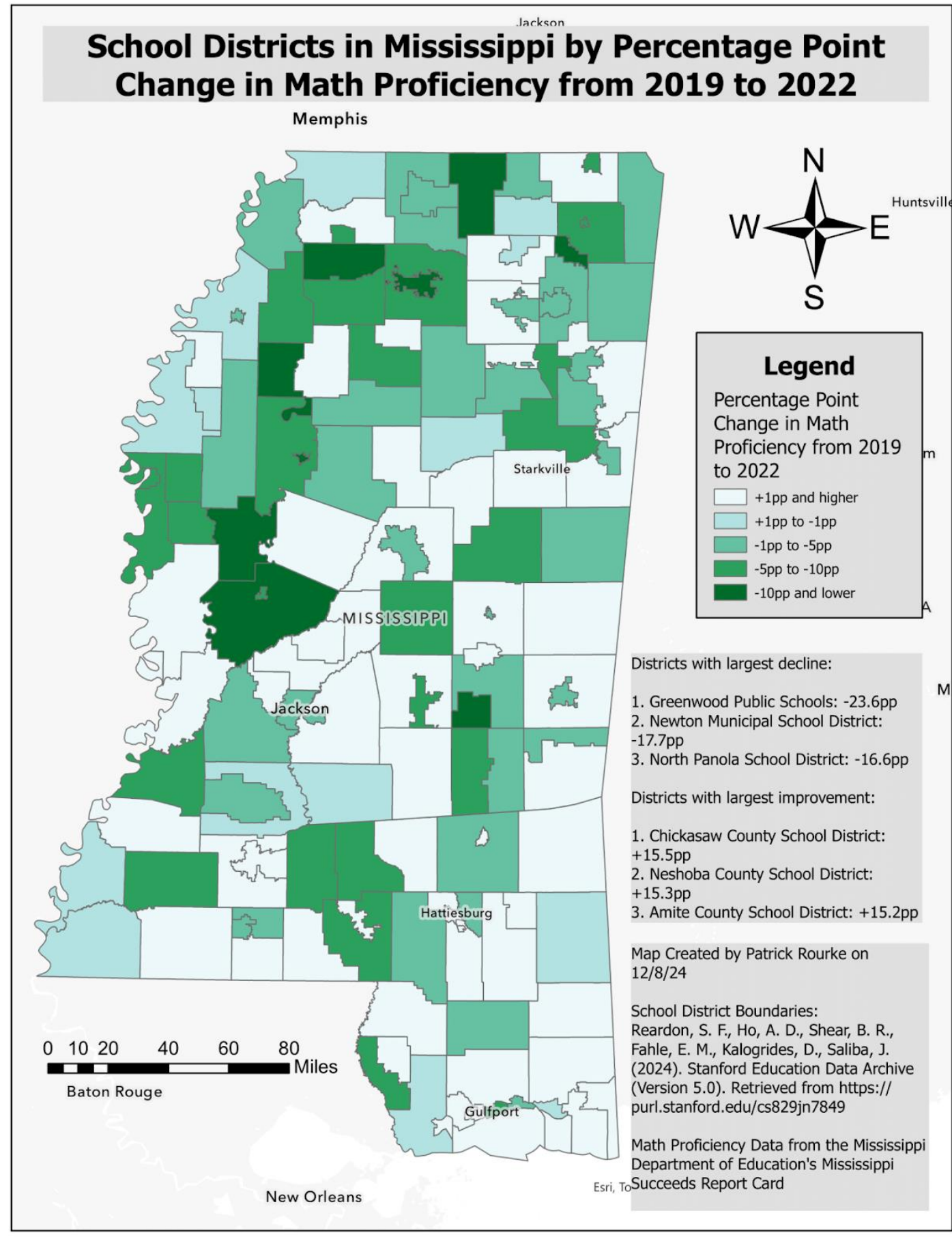
Percentage of Population with No Internet Access



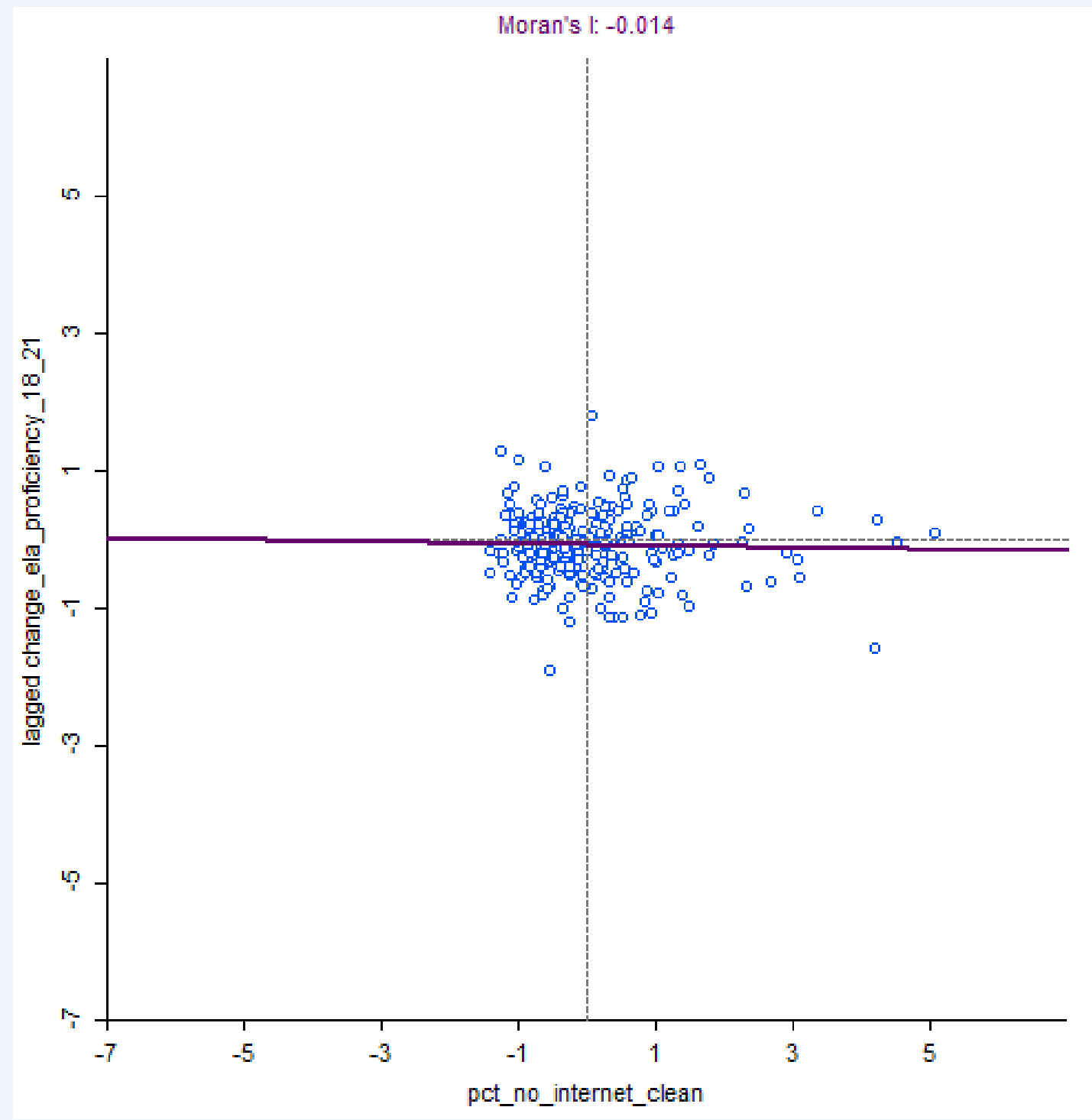
Percentage Point Change in ELA Proficiency



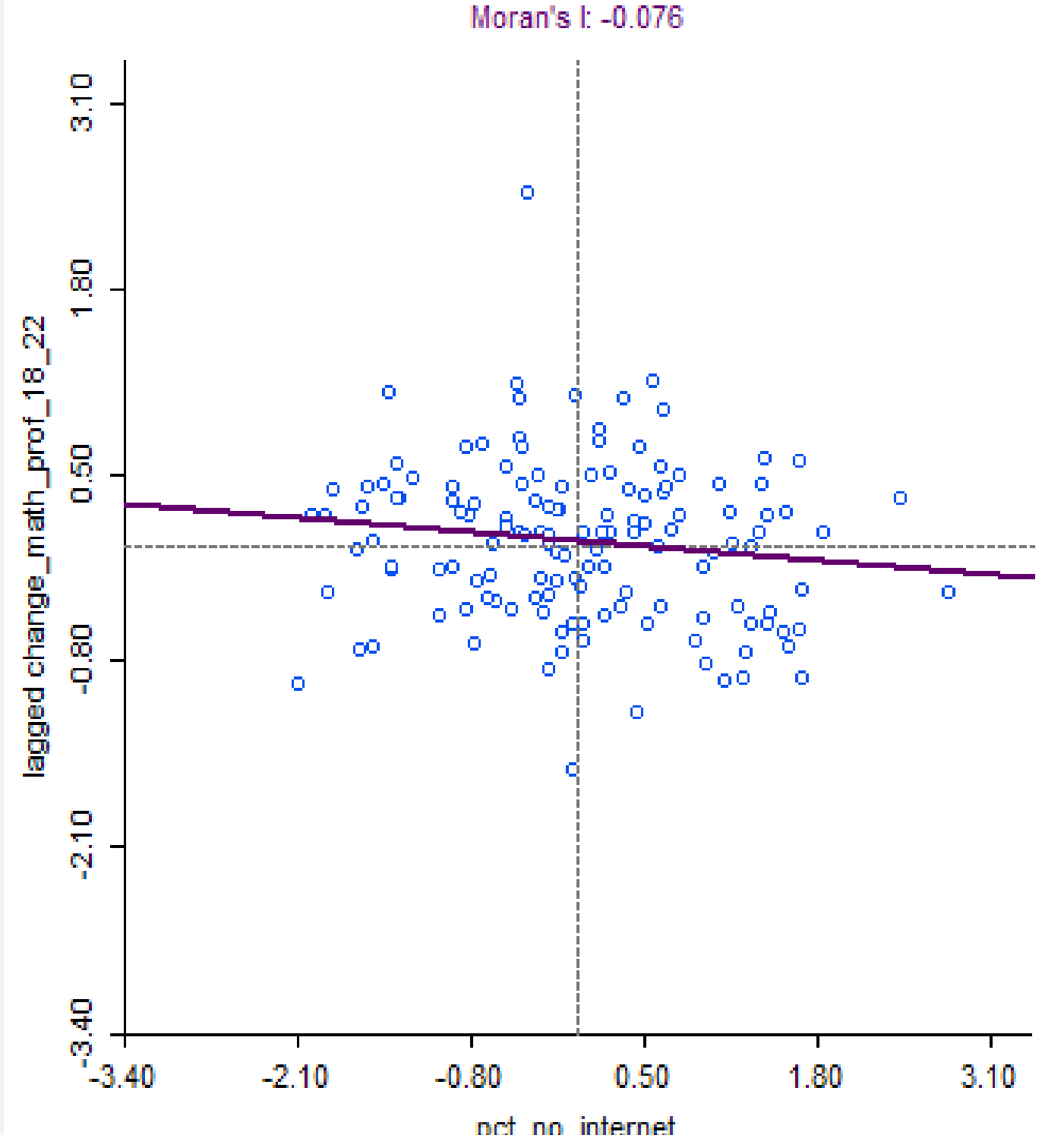
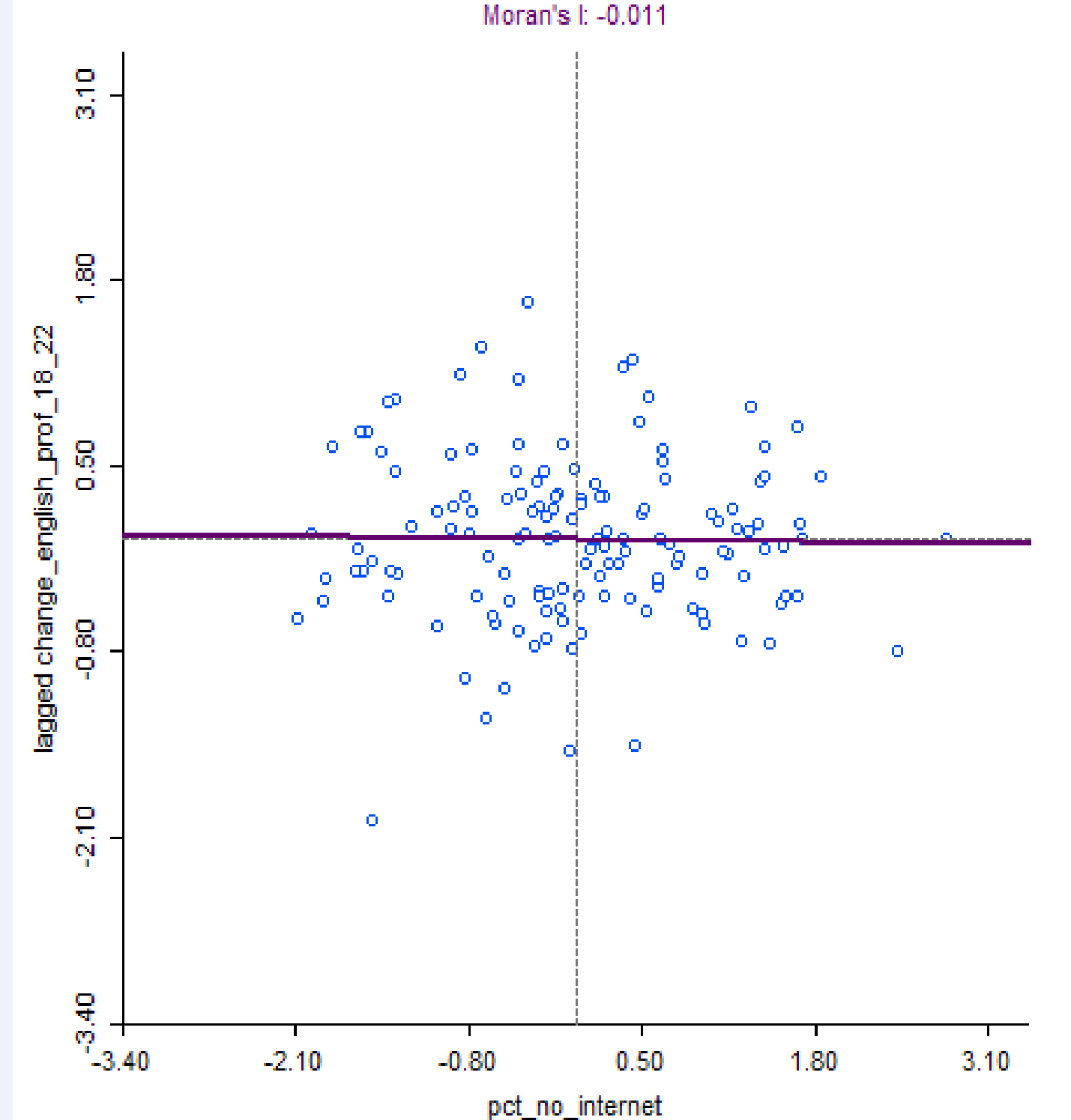
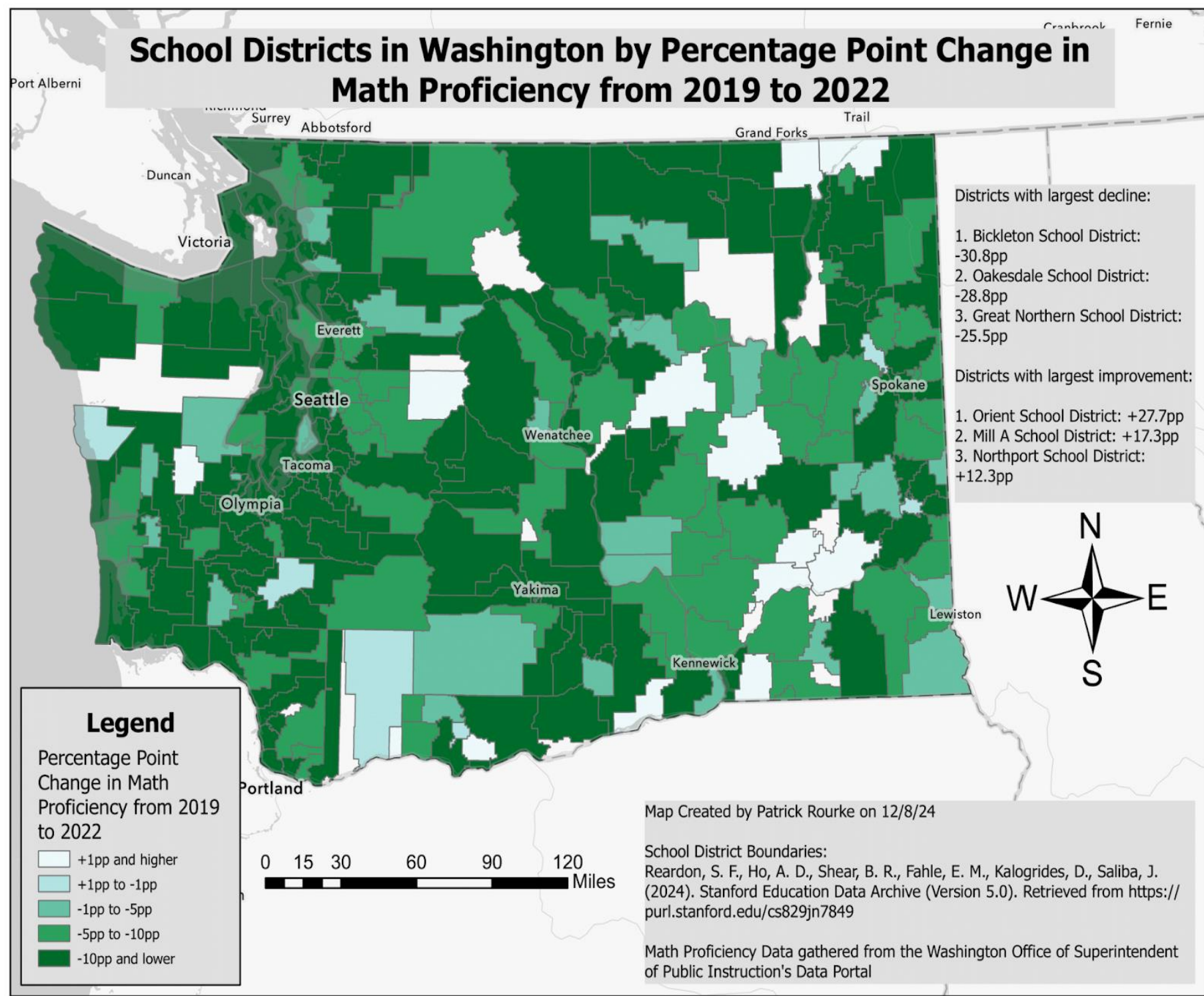
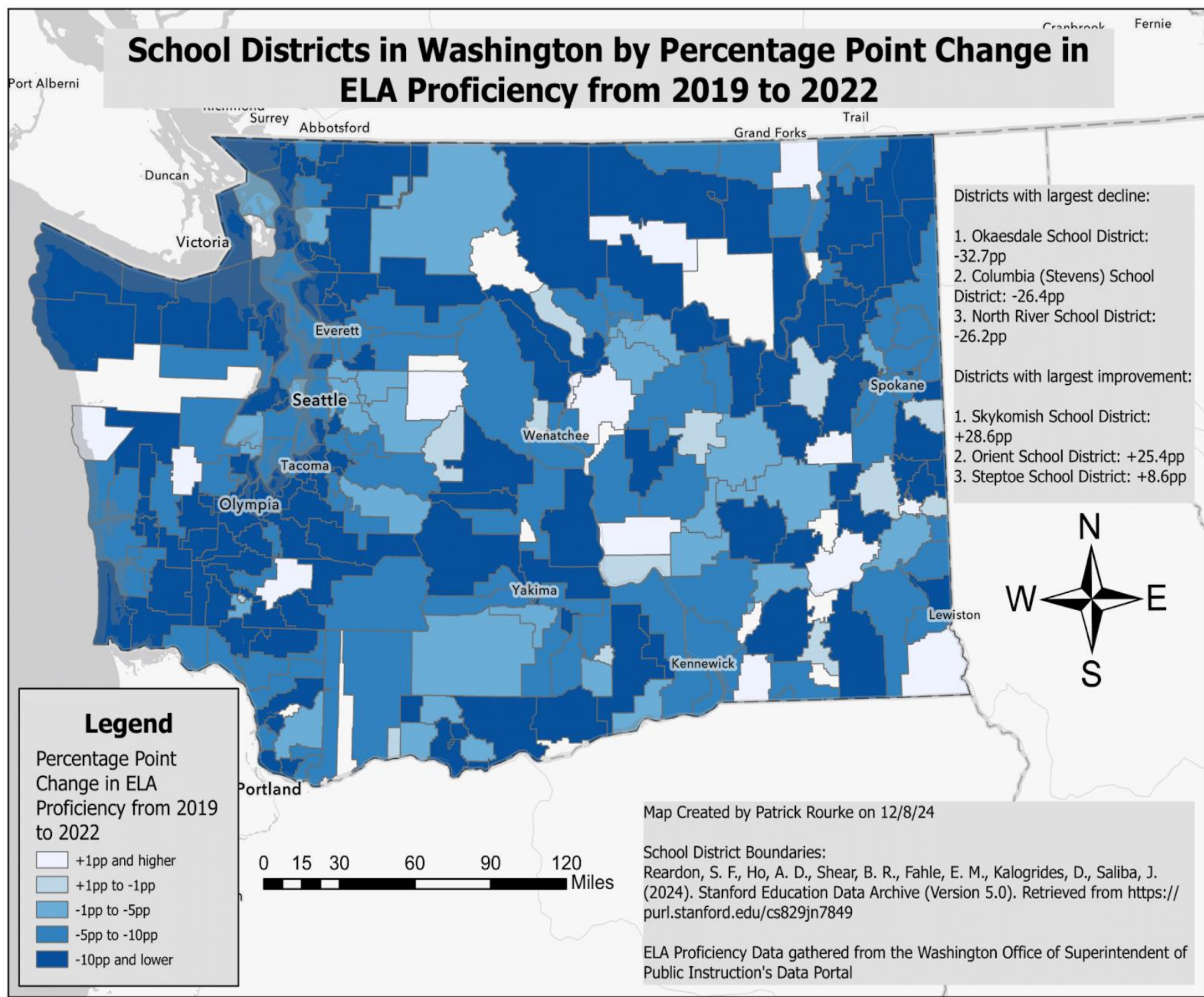
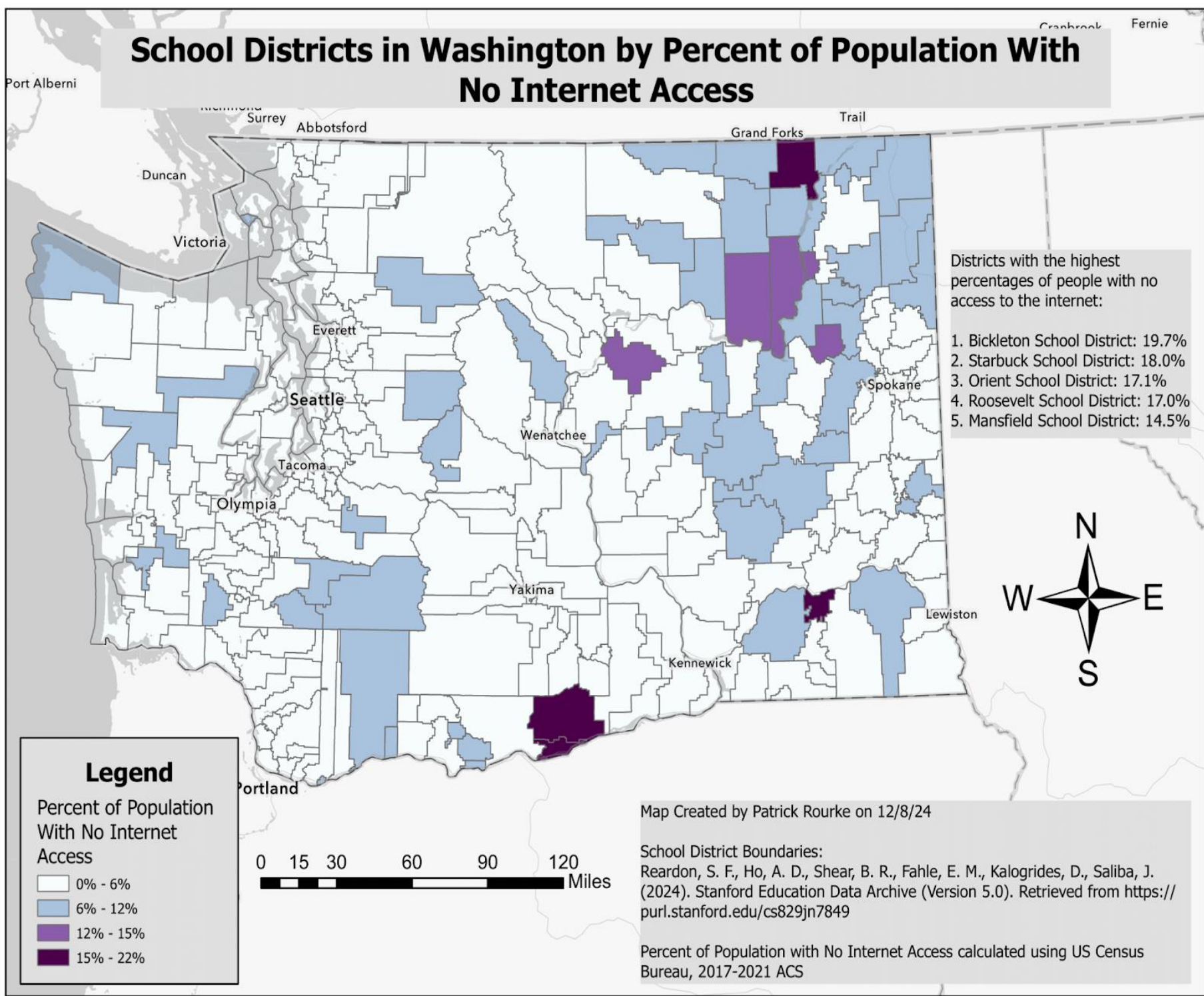
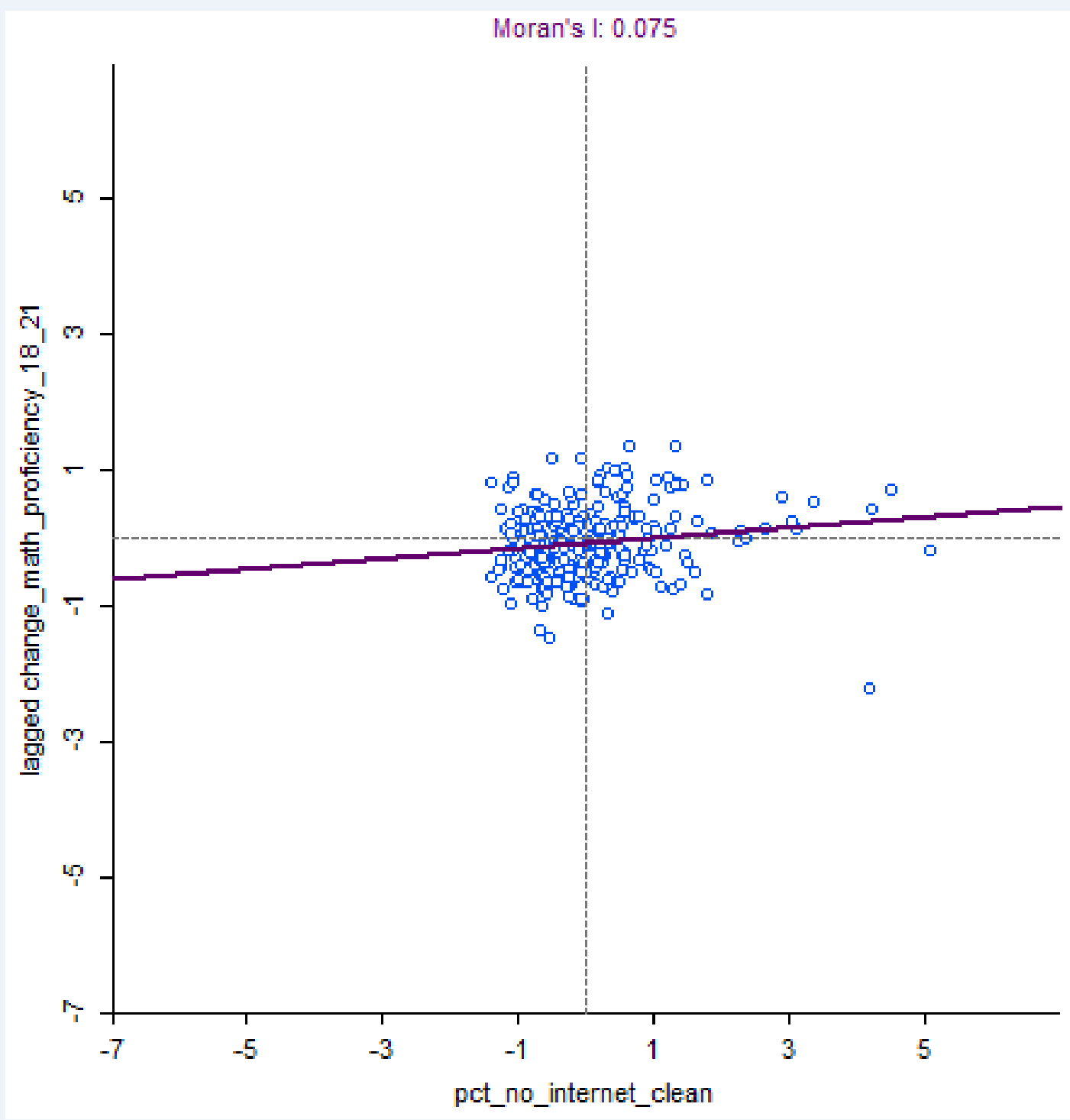
Percentage Point Change in Math Proficiency



Bivariate Moran's I (change in ELA proficiency by % with no internet access)



Bivariate Moran's I (change in Math proficiency by % with no internet access)



GIS Methods Overview

- Step 1: Data Cleaning**
- Step 2: Local Bivariate Relationship (in ArcGIS)**
 - Used to visualize the relationship between two variables on a map by calculating if the values of one variable are influenced by the values of another.
- Step 3: Bivariate Moran's I (in GeoDa)**
 - Used to further test if there is a spatial relationship between areas with low levels of internet access and areas with larger decreases in test score proficiency.
- Step 4: Spatial Regression (in GeoDa)**
 - Used to further test if there is a relationship between percentages of the population with no internet access and changes in test score proficiency.

Results

- Local Bivariate Relationship**
 - Only significant for some districts in Washington; not included on poster due to overwhelmingly insignificant results
- Bivariate Moran's I**
 - None of these results were significant (to have been significant, the Moran's I value would have had to be greater than 0.3).
 - Since none of the results were significant, we cannot conclude that neighboring school districts tend to have similar or dissimilar values for the change in test score proficiency from 2019 to 2022 and percentage of the population with no internet access
- Spatial Regression**
 - Results were not significant; not included on poster due to insignificant results.

Discussion

- My analysis suggests that there is no relationship between lack of internet access and the change in test score proficiency seen from 2019 to 2022.
- However, internet access may still be an important factor that impacted an individual student's ability to score highly on tests immediately after the pandemic. *Here are some things to keep in mind:*
 - Other variables (ex. Race, SES, etc.) could affect the change in test scores
 - Different states have different proficiency standards and tests
 - Results may differ in the other 48 states
- More research on this topic is needed to conclude whether internet access had an impact on the changes in test score proficiency seen between 2019 and 2022.

References

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